MATLAB Program

```
% Mass-Spring-Damper System Simulation
clc;
clear;
close all;

% System Parameters
m = 2;      % Mass (kg)
c = 5;      % Damping coefficient (Ns/m)
k = 25;     % Spring stiffness (N/m)

% Transfer Function
num = [1];
den = [m c k];

sys = tf(num, den);

disp('Transfer Function:');
sys

% Step Response
figure;
step(sys);
grid on;
title('Step Response of Mass-Spring-Damper System');
```

```
% Performance Parameters
info = stepinfo(sys);
disp(info);

% Effect of Different Damping Values
figure;
hold on;
c_values = [0.5 2 8];
for c = c_values
    sys = tf([1],[m c k]);
    step(sys);
end
legend('c=0.5','c=2','c=8');
title('Effect of Damping Coefficient');
grid on;
```

Simulink

